Outputs do Programa Econométrico

Pass-through cambial e política monetária no Brasil — VAR/VECM

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1. Base e transformações

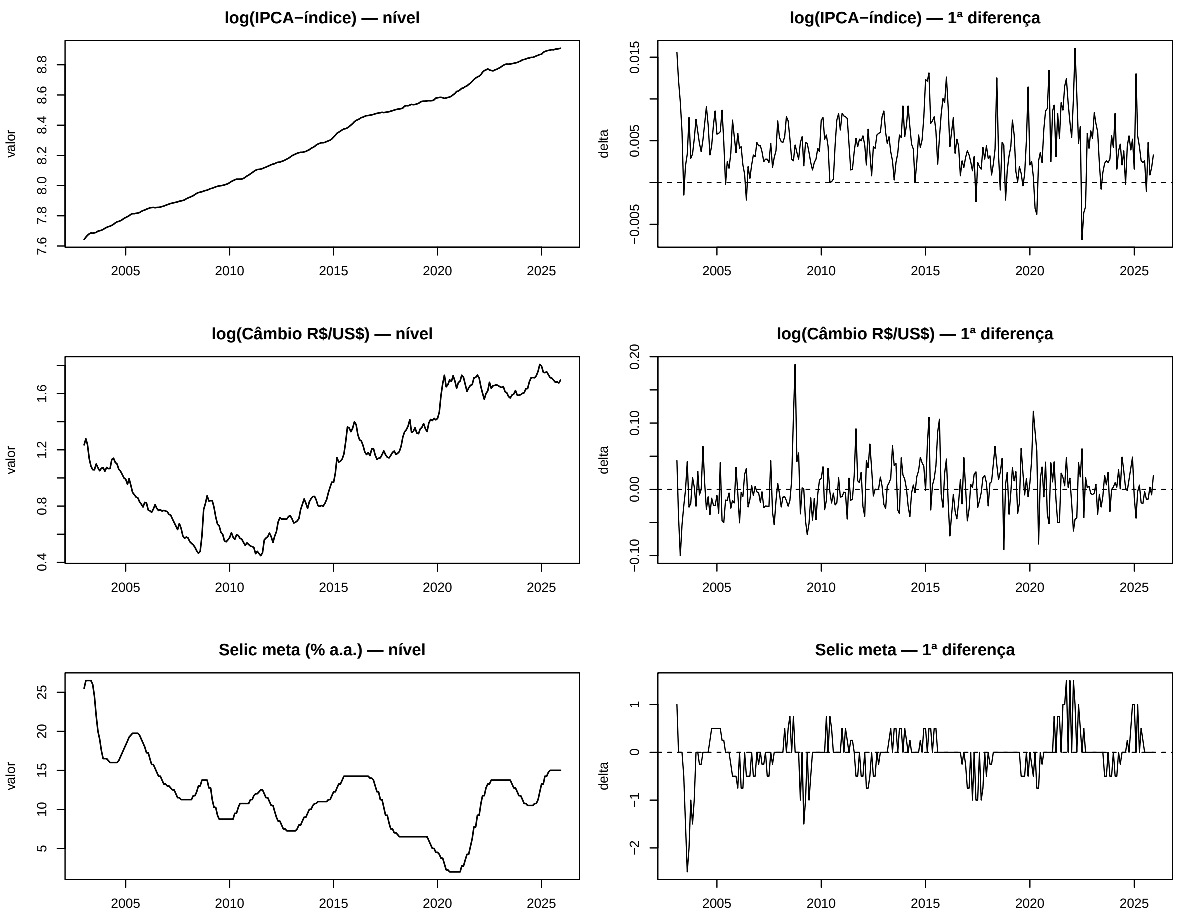
```
## Observações: 276
```

```
## Período: 2003-01-01 a 2025-12-01
```

Table 1: Estatísticas descritivas das séries transformadas

variavel	n	media	dp	min	max
log__ipca__indice	276	8.2916	0.3709	7.6429	8.9097
log__cambio	276	1.1050	0.4130	0.4472	1.8078
selic	276	11.6141	4.6527	2.0000	26.5000

2. Gráficos em nível e primeira diferença



3. Testes ADF

Table 2: Teste ADF — séries em nível e 1ª diferença

Serie	Tipo	Estatistica	CV 1%	CV 5%	CV 10%	Rejeita H0 5%
log(IPCA) — nível (trend)	trend	-2.3088	-3.98	-3.42	-3.13	não
log(Câmbio) — nível (trend)	trend	-3.0604	-3.98	-3.42	-3.13	não
Selic — nível (trend)	trend	-2.6863	-3.98	-3.42	-3.13	não
Selic — nível (drift; complementar)	drift	-3.5339	-3.44	-2.87	-2.57	SIM
log(IPCA) — 1ª dif.	drift	-7.8331	-3.44	-2.87	-2.57	SIM
log(Câmbio) — 1ª dif.	drift	-10.2484	-3.44	-2.87	-2.57	SIM
Selic — 1ª dif.	drift	-5.2714	-3.44	-2.87	-2.57	SIM

```
## --- ADF log(IPCA) nível ---

##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0118224 -0.0012990 -0.0001148  0.0013751  0.0099914
##
## Coefficients:
##                Estimate  Std. Error t value            Pr(>|t|)
## (Intercept)  0.14067773  0.06002665   2.344              0.0198 *
## z.lag.1     -0.01812400  0.00785009  -2.309              0.0217 *
## tt           0.00008381  0.00003649   2.297              0.0224 *
## z.diff.lag   0.55499080  0.04933659  11.249 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.00271 on 270 degrees of freedom
## Multiple R-squared:  0.3232, Adjusted R-squared:  0.3157
## F-statistic: 42.98 on 3 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -2.3088 19.7487 2.6755
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau3 -3.98 -3.42 -3.13
## phi2  6.15  4.71  4.05
## phi3  8.34  6.30  5.36

##
## --- ADF log(Câmbio) nível ---
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.106478 -0.019884 -0.001079  0.016433  0.149839
##
## Coefficients:
##                 Estimate  Std. Error t value     Pr(>|t|)
## (Intercept)  0.00796256  0.00586609   1.357     0.175792
## z.lag.1     -0.02429806  0.00793962  -3.060     0.002433 **
## tt           0.00014321  0.00004156   3.446     0.000659 ***
## z.diff.lag   0.31930064  0.05640949   5.660 0.0000000385 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03329 on 270 degrees of freedom
## Multiple R-squared:  0.1514, Adjusted R-squared:  0.142
## F-statistic: 16.06 on 3 and 270 DF,  p-value: 0.000000001231
##
##
## Value of test-statistic is: -3.0604 4.1445 6.097
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau3 -3.98 -3.42 -3.13
## phi2  6.15  4.71  4.05
## phi3  8.34  6.30  5.36

##
## --- ADF Selic nível - trend ---

##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.45590 -0.20166  0.01476  0.22503  1.45859
```

```
##
## Coefficients:
##               Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.1282078  0.1079345   1.188              0.23594
## z.lag.1     -0.0164489  0.0061232  -2.686              0.00767 **
## tt           0.0002854  0.0003621   0.788              0.43132
## z.diff.lag   0.5142022  0.0507784  10.126 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4023 on 270 degrees of freedom
## Multiple R-squared:  0.3279, Adjusted R-squared:  0.3204
## F-statistic:  43.9 on 3 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -2.6863 4.619 6.5461
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau3 -3.98 -3.42 -3.13
## phi2  6.15  4.71  4.05
## phi3  8.34  6.30  5.36
```

```
##
## --- ADF Selic nível - drift complementar ---
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.45548 -0.20061  0.00923  0.22511  1.47864
##
## Coefficients:
##              Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.195483   0.066009   2.961             0.003333 **
## z.lag.1     -0.018824   0.005327  -3.534             0.000481 ***
## z.diff.lag   0.522384   0.049671  10.517 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.402 on 271 degrees of freedom
## Multiple R-squared:  0.3263, Adjusted R-squared:  0.3213
## F-statistic: 65.63 on 2 and 271 DF,  p-value: < 0.00000000000000022
##
##
```

```
## Value of test-statistic is: -3.5339 6.6272
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

```
##
## --- ADF log(IPCA) 1ª diferença ---
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0124555 -0.0014312 -0.0000663  0.0014531  0.0099951
##
## Coefficients:
##               Estimate Std. Error t value         Pr(>|t|)
## (Intercept)  0.0019964  0.0003087   6.467 0.00000000046657 ***
## z.lag.1     -0.4442931  0.0567203  -7.833 0.00000000000011 ***
## z.diff.lag  -0.0378421  0.0596315  -0.635            0.526
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002732 on 270 degrees of freedom
## Multiple R-squared:  0.2365, Adjusted R-squared:  0.2308
## F-statistic: 41.81 on 2 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -7.8331 30.716
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

```
##
## --- ADF log(Câmbio) 1ª diferença ---
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
```

```
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.107331 -0.021145 -0.001864  0.019388  0.147624
##
## Coefficients:
##              Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  0.001277   0.002045   0.624               0.533
## z.lag.1     -0.711781   0.069453 -10.248 <0.0000000000000002 ***
## z.diff.lag   0.081793   0.060247   1.358               0.176
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03374 on 270 degrees of freedom
## Multiple R-squared:  0.3344, Adjusted R-squared:  0.3294
## F-statistic: 67.82 on 2 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -10.2484 52.5181
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

```
##
## --- ADF Selic 1ª diferença ---
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.78815 -0.23677  0.01323  0.19296  0.94356
##
## Coefficients:
##             Estimate Std. Error t value        Pr(>|t|)
## (Intercept) -0.01323    0.02305  -0.574           0.566
## z.lag.1     -0.28107    0.05332  -5.271 0.0000002775515 ***
## z.diff.lag  -0.37978    0.05556  -6.835 0.0000000000544 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3792 on 270 degrees of freedom
## Multiple R-squared:  0.3405, Adjusted R-squared:  0.3356
## F-statistic:  69.7 on 2 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -5.2714 13.897
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

4. Testes Zivot-Andrews

Table 3: Teste Zivot-Andrews — séries em nível

Serie	Estatistica	CV 5%	Quebra	Rejeita H0 5%
log(IPCA) — nível	-2.4927	-5.08	2014-11-01	não
log(Câmbio) — nível	-3.2891	-5.08	2004-06-01	não
Selic — nível	-3.4649	-5.08	2017-05-01	não

Table 4: Teste Zivot-Andrews — séries em 1ª diferença

Serie	Estatistica	CV 5%	Quebra	Rejeita H0 5%
log(IPCA) — 1ª dif.	-9.7295	-5.08	2020-08-01	SIM
log(Câmbio) — 1ª dif.	-12.2660	-5.08	2015-09-01	SIM
Selic — 1ª dif.	-10.5953	-5.08	2003-09-01	SIM

```
## --- ZA log(IPCA) nível ---

##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0099043 -0.0020648 -0.0001453  0.0017957  0.0123271
##
## Coefficients:
##               Estimate  Std. Error t value             Pr(>|t|)
## (Intercept)  0.23038758  0.09047986   2.546              0.01144 *
## y.l1         0.97055094  0.01181419  82.151 < 0.0000000000000002 ***
```

```
## trend        0.00012547  0.00005198   2.414              0.01646 *
## du           0.00261111  0.00098722   2.645              0.00865 **
## dt          -0.00001146  0.00001005  -1.140              0.25542
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.003298 on 270 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9999
## F-statistic: 8.598e+05 on 4 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -2.4927
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 143
```

```
##
## --- ZA log(Câmbio) nível ---
```

```
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.092122 -0.021251 -0.002381  0.016906  0.186283
##
## Coefficients:
##               Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  0.0003227  0.0228307   0.014              0.9887
## y.l1         0.9665742  0.0101626  95.111 <0.0000000000000002 ***
## trend        0.0031182  0.0017353   1.797              0.0735 .
## du          -0.0461010  0.0177992  -2.590              0.0101 *
## dt          -0.0028969  0.0017390  -1.666              0.0969 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.035 on 270 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.9929, Adjusted R-squared:  0.9928
## F-statistic:  9503 on 4 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -3.2891
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 18
```

```
##
## --- ZA Selic nível ---
```

```
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.03044 -0.20106  0.00345  0.20660  1.50327
##
## Coefficients:
##               Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.2638672  0.1692007   1.559             0.120051
## y.l1         0.9698696  0.0086960 111.531 < 0.0000000000000002 ***
## trend        0.0005956  0.0008371   0.712             0.477369
## du          -0.4577781  0.1240581  -3.690             0.000271 ***
## dt           0.0069923  0.0021066   3.319             0.001027 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4659 on 270 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.9898, Adjusted R-squared:  0.9897
## F-statistic:  6565 on 4 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -3.4649
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 173
```

```
##
## --- ZA log(IPCA) 1ª diferença ---
```

```
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0131876 -0.0013075 -0.0000279  0.0014849  0.0107931
##
```

```
## Coefficients:
##                 Estimate   Std. Error t value            Pr(>|t|)
## (Intercept)  0.002423056  0.000462698   5.237          0.00000033 ***
## y.l1         0.504921286  0.050884489   9.923 < 0.0000000000000002 ***
## trend       -0.000002511  0.000003096  -0.811             0.41808
## du           0.002269163  0.000806769   2.813             0.00528 **
## dt          -0.000048362  0.000018984  -2.548             0.01141 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002699 on 269 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.3311, Adjusted R-squared:  0.3212
## F-statistic: 33.29 on 4 and 269 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -9.7295
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 211

##
## --- ZA log(Câmbio) 1ª diferença ---

##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.105552 -0.020289 -0.001054  0.017536  0.157475
##
## Coefficients:
##                Estimate  Std. Error t value    Pr(>|t|)
## (Intercept) -0.01583646  0.00562304  -2.816     0.00522 **
## y.l1         0.29304591  0.05763506   5.085 0.000000691 ***
## trend        0.00021237  0.00006395   3.321     0.00102 **
## du          -0.01664722  0.00816259  -2.039     0.04238 *
## dt          -0.00018142  0.00010643  -1.705     0.08942 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03341 on 269 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.1487, Adjusted R-squared:  0.136
## F-statistic: 11.74 on 4 and 269 DF,  p-value: 0.000000008317
##
##
## Teststatistic: -12.266
```

```
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 152
```

```
##
## --- ZA Selic 1ª diferença ---
```

```
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.45736 -0.21789 -0.00407  0.20800  1.46279
##
## Coefficients:
##             Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  0.42063    0.41219   1.020             0.30842
## y.l1         0.44147    0.05271   8.375 0.00000000000000309 ***
## trend       -0.22569    0.07855  -2.873             0.00439 **
## du           1.30783    0.28737   4.551 0.00000809320262047 ***
## dt           0.22622    0.07857   2.879             0.00431 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3915 on 269 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.3657, Adjusted R-squared:  0.3563
## F-statistic: 38.77 on 4 and 269 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -10.5953
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 8
```

5. Engle-Granger

Table 5: Teste de Engle-Granger — ADF nos resíduos com valores críticos de MacKinnon

Variável dependente	Estatistica	CV 1% MacKinnon	CV 5% MacKinnon	CV 10% MacKinnon	Rejeita H0 5%
log(IPCA)	-2.0427	-4.29	-3.74	-3.45	não
log(Câmbio)	-2.5580	-4.29	-3.74	-3.45	não
Selic	-2.2761	-4.29	-3.74	-3.45	não

```
## --- Regressão EG: log(IPCA) dependente ---

##
## Call:
## lm(formula = log_ipca_indice ~ log_cambio + selic, data = Y)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.44839 -0.10737  0.02722  0.14369  0.33126
##
## Coefficients:
##              Estimate Std. Error t value           Pr(>|t|)
## (Intercept)  7.885361   0.045475  173.40 <0.0000000000000002 ***
## log_cambio   0.665192   0.027178   24.48 <0.0000000000000002 ***
## selic       -0.028313   0.002413  -11.73 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.184 on 273 degrees of freedom
## Multiple R-squared:  0.7558, Adjusted R-squared:  0.754
## F-statistic: 422.4 on 2 and 273 DF,  p-value: < 0.00000000000000022

##
## --- ADF nos resíduos: log(IPCA) dependente ---

##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.098722 -0.010978  0.002015  0.018200  0.085525
##
## Coefficients:
##             Estimate Std. Error t value          Pr(>|t|)
## z.lag.1    -0.017592   0.008612  -2.043             0.042 *
## z.diff.lag  0.424239   0.054869   7.732 0.000000000000206 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02579 on 272 degrees of freedom
## Multiple R-squared:  0.1854, Adjusted R-squared:  0.1794
## F-statistic: 30.96 on 2 and 272 DF,  p-value: 0.0000000000007712
##
##
## Value of test-statistic is: -2.0427
```

```
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
```

```
##
## --- Regressão EG: log(Câmbio) dependente ---
```

```
##
## Call:
## lm(formula = log_cambio ~ log_ipca_indice + selic, data = Y)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49160 -0.16533 -0.02697  0.09702  0.54518
##
## Coefficients:
##                  Estimate Std. Error t value            Pr(>|t|)
## (Intercept)     -7.747737   0.370033 -20.938 < 0.0000000000000002 ***
## log_ipca_indice  1.032697   0.042193  24.475 < 0.0000000000000002 ***
## selic            0.024975   0.003363   7.426     0.00000000000143 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2292 on 273 degrees of freedom
## Multiple R-squared:  0.6943, Adjusted R-squared:  0.6921
## F-statistic:   310 on 2 and 273 DF,  p-value: < 0.00000000000000022
```

```
##
## --- ADF nos resíduos: log(Câmbio) dependente ---
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.106824 -0.025270 -0.003154  0.015630  0.144776
##
## Coefficients:
##             Estimate Std. Error t value        Pr(>|t|)
## z.lag.1    -0.024217   0.009467  -2.558          0.0111 *
## z.diff.lag  0.379435   0.055684   6.814 0.0000000000609 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.03553 on 272 degrees of freedom
## Multiple R-squared:  0.1591, Adjusted R-squared:  0.1529
## F-statistic: 25.73 on 2 and 272 DF,  p-value: 0.00000000005844
##
##
## Value of test-statistic is: -2.558
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
```

```
##
## --- Regressão EG: Selic dependente ---
```

```
##
## Call:
## lm(formula = selic ~ log_ipca_indice + log_cambio, data = Y)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1352  -2.3500   0.2556   2.6453   7.8621
##
## Coefficients:
##                 Estimate Std. Error t value             Pr(>|t|)
## (Intercept)     102.3721     7.5988  13.472 < 0.0000000000000002 ***
## log_ipca_indice -11.8426     1.0092 -11.735 < 0.0000000000000002 ***
## log_cambio        6.7289     0.9061   7.426    0.00000000000143 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.762 on 273 degrees of freedom
## Multiple R-squared:  0.3509, Adjusted R-squared:  0.3462
## F-statistic: 73.81 on 2 and 273 DF,  p-value: < 0.00000000000000022
```

```
##
## --- ADF nos resíduos: Selic dependente ---
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8560 -0.2272  0.0304  0.2650  1.9108
##
## Coefficients:
```

```
##             Estimate Std. Error t value           Pr(>|t|)
## z.lag.1    -0.017421   0.007654  -2.276             0.0236 *
## z.diff.lag  0.530698   0.051185  10.368 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4693 on 272 degrees of freedom
## Multiple R-squared:  0.2866, Adjusted R-squared:  0.2814
## F-statistic: 54.65 on 2 and 272 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -2.2761
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
```

6. Johansen

```
## Defasagem selecionada pelo AIC para Johansen: K = 5
```

```
## Rank pelo traço a 5%: 2
```

```
## Rank pelo máximo autovalor a 5%: 2
```

Table 6: Johansen — teste por traço

Teste	Hipotese	Estatistica	CV 10%	CV 5%	CV 1%	Rejeita H0 5%
Traço	r <= 2 |	5.5625	7.52	9.24	12.97	não
Traço	r <= 1 |	22.6679	17.85	19.96	24.60	SIM
Traço	r = 0 |	75.8709	32.00	34.91	41.07	SIM

Table 7: Johansen — teste por máximo autovalor

Teste	Hipotese	Estatistica	CV 10%	CV 5%	CV 1%	Rejeita H0 5%
Máximo autovalor	r <= 2 |	5.5625	7.52	9.24	12.97	não
Máximo autovalor	r <= 1 |	17.1054	13.75	15.67	20.20	SIM
Máximo autovalor	r = 0 |	53.2030	19.77	22.00	26.81	SIM

```
## --- Seleção de defasagem para VAR/Johansen ---
```

```
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      5      4      4      5
```

```
##
## --- Johansen por traço ---
```

```
##
## ######################
## # Johansen-Procedure #
## ######################
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1]  0.1782515234542451088018  0.0611686646270932221636
## [3]  0.0203167173182916549479 -0.0000000000000000154533
##
## Values of teststatistic and critical values of test:
##
##           test 10pct  5pct  1pct
## r <= 2 |  5.56  7.52  9.24 12.97
## r <= 1 | 22.67 17.85 19.96 24.60
## r = 0  | 75.87 32.00 34.91 41.07
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##                    log_ipca_indice.l5 log_cambio.l5    selic.l5    constant
## log_ipca_indice.l5         1.00000000     1.0000000  1.00000000  1.00000000
## log_cambio.l5             -0.50313682    -0.3766748 -0.76616601  0.97110859
## selic.l5                  -0.03251781     0.4222753  0.02255715  0.04079072
## constant                  -9.52373960   -13.3112620 -7.71344432 -9.73089269
##
## Weights W:
## (This is the loading matrix)
##
##                   log_ipca_indice.l5 log_cambio.l5     selic.l5
## log_ipca_indice.d      -0.0012309951 -0.0001609359 -0.000664139
## log_cambio.d           -0.0006973477 -0.0018796890  0.024889249
## selic.d                 0.0729031082 -0.0442949400 -0.089948017
##                                   constant
## log_ipca_indice.d -0.00000000000000005672475
## log_cambio.d       0.00000000000000005467947
## selic.d           -0.00000000000002617301619

##
## --- Johansen por máximo autovalor ---

##
## ######################
## # Johansen-Procedure #
## ######################
##
## Test type: maximal eigenvalue statistic (lambda max) , without linear trend and constant in cointegra
##
## Eigenvalues (lambda):
## [1]  0.1782515234542451088018  0.0611686646270932221636
## [3]  0.0203167173182916549479 -0.0000000000000000154533
##
## Values of teststatistic and critical values of test:
```

```
##
##           test 10pct  5pct  1pct
## r <= 2 |  5.56  7.52  9.24 12.97
## r <= 1 | 17.11 13.75 15.67 20.20
## r = 0  | 53.20 19.77 22.00 26.81
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##                    log_ipca_indice.l5 log_cambio.l5    selic.l5    constant
## log_ipca_indice.l5         1.00000000     1.0000000  1.00000000  1.00000000
## log_cambio.l5             -0.50313682    -0.3766748 -0.76616601  0.97110859
## selic.l5                  -0.03251781     0.4222753  0.02255715  0.04079072
## constant                  -9.52373960   -13.3112620 -7.71344432 -9.73089269
##
## Weights W:
## (This is the loading matrix)
##
##                   log_ipca_indice.l5 log_cambio.l5     selic.l5
## log_ipca_indice.d      -0.0012309951 -0.0001609359 -0.000664139
## log_cambio.d           -0.0006973477 -0.0018796890  0.024889249
## selic.d                 0.0729031082 -0.0442949400 -0.089948017
##                                    constant
## log_ipca_indice.d -0.000000000000000005672475
## log_cambio.d       0.000000000000000005467947
## selic.d           -0.000000000000002617301619
```

7. VAR em primeira diferença

```
## Ordem das variáveis no VAR: log_cambio -> log_ipca_indice -> selic
```

```
## Defasagem selecionada pelo AIC: p = 4
```

```
## Critérios de informação:
```

```
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      4      3      3      4
```

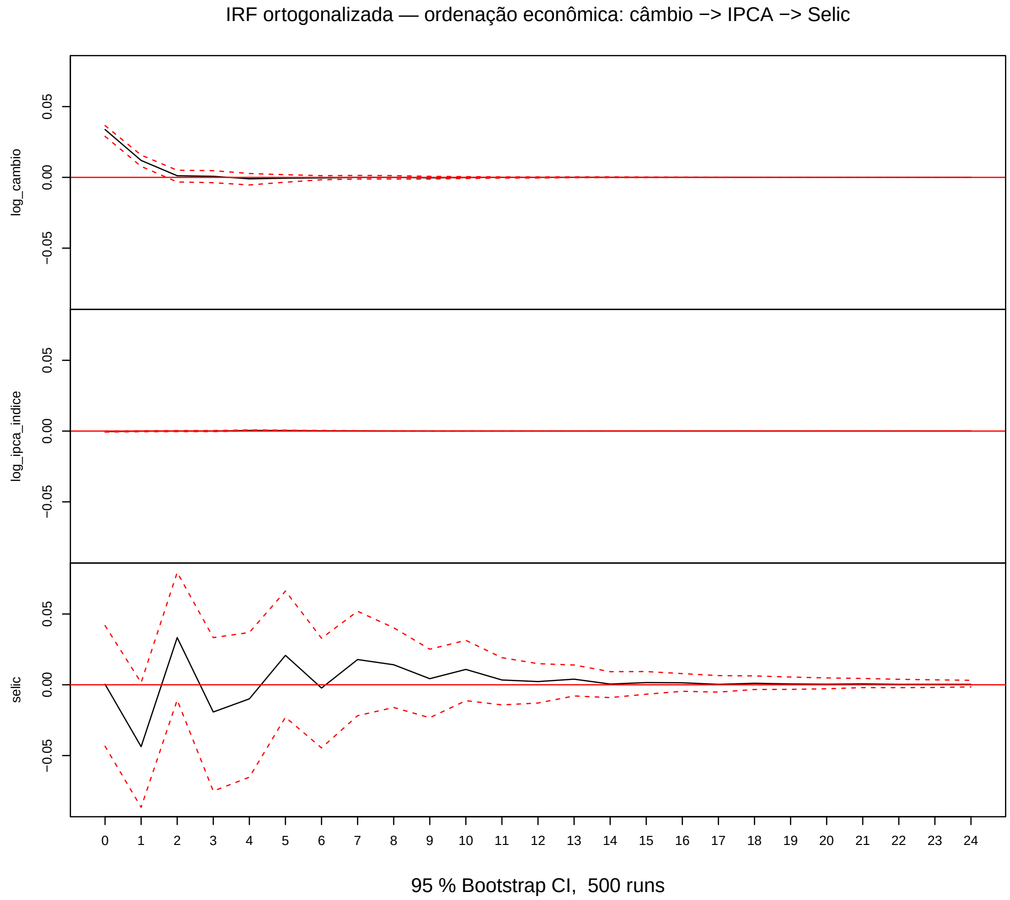
```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: log_cambio, log_ipca_indice, selic
## Deterministic variables: const
## Sample size: 271
## Log Likelihood: 1674.421
## Roots of the characteristic polynomial:
## 0.8202 0.7444 0.7444 0.6874 0.6874 0.5953 0.5953 0.554 0.554 0.5126 0.5126 0.4513
## Call:
## VAR(y = dY_econ, p = p_var, type = "const")
##
##
## Estimation results for equation log_cambio:
```

```
## ============================================
## log_cambio = log_cambio.l1 + log_ipca_indice.l1 + selic.l1 + log_cambio.l2 + log_ipca_indice.l2 + sel
##
##                      Estimate Std. Error t value    Pr(>|t|)
## log_cambio.l1       0.3549433  0.0624154   5.687 0.000000035 ***
## log_ipca_indice.l1  0.1101646  0.7707920   0.143       0.886
## selic.l1            0.0030420  0.0058931   0.516       0.606
## log_cambio.l2      -0.0927327  0.0662833  -1.399       0.163
## log_ipca_indice.l2 -0.5869891  0.8633510  -0.680       0.497
## selic.l2           -0.0018104  0.0057772  -0.313       0.754
## log_cambio.l3       0.0441540  0.0669723   0.659       0.510
## log_ipca_indice.l3  0.8514769  0.8659203   0.983       0.326
## selic.l3            0.0053416  0.0056692   0.942       0.347
## log_cambio.l4      -0.0482857  0.0623085  -0.775       0.439
## log_ipca_indice.l4 -1.0862145  0.7709851  -1.409       0.160
## selic.l4           -0.0006228  0.0056040  -0.111       0.912
## const               0.0052755  0.0050486   1.045       0.297
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.0338 on 258 degrees of freedom
## Multiple R-Squared: 0.1262,  Adjusted R-squared: 0.08559
## F-statistic: 3.106 on 12 and 258 DF,  p-value: 0.0003929
##
##
## Estimation results for equation log_ipca_indice:
## ===================================================
## log_ipca_indice = log_cambio.l1 + log_ipca_indice.l1 + selic.l1 + log_cambio.l2 + log_ipca_indice.l2
##
##                       Estimate  Std. Error t value          Pr(>|t|)
## log_cambio.l1       0.00204292  0.00500775   0.408            0.6836
## log_ipca_indice.l1  0.48623754  0.06184270   7.862 0.000000000000103 ***
## selic.l1            0.00068490  0.00047282   1.449            0.1487
## log_cambio.l2       0.00147842  0.00531808   0.278            0.7812
## log_ipca_indice.l2  0.05069004  0.06926896   0.732            0.4650
## selic.l2            0.00048952  0.00046352   1.056            0.2919
## log_cambio.l3      -0.00073324  0.00537337  -0.136            0.8916
## log_ipca_indice.l3 -0.02461471  0.06947510  -0.354            0.7234
## selic.l3            0.00008533  0.00045485   0.188            0.8513
## log_cambio.l4       0.01248351  0.00499918   2.497            0.0131 *
## log_ipca_indice.l4 -0.07670887  0.06185820  -1.240            0.2161
## selic.l4           -0.00043537  0.00044963  -0.968            0.3338
## const               0.00256194  0.00040506   6.325 0.000000001111975 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.002712 on 258 degrees of freedom
## Multiple R-Squared: 0.3323,  Adjusted R-squared: 0.3012
## F-statistic:  10.7 on 12 and 258 DF,  p-value: < 0.00000000000000022
##
##
## Estimation results for equation selic:
```

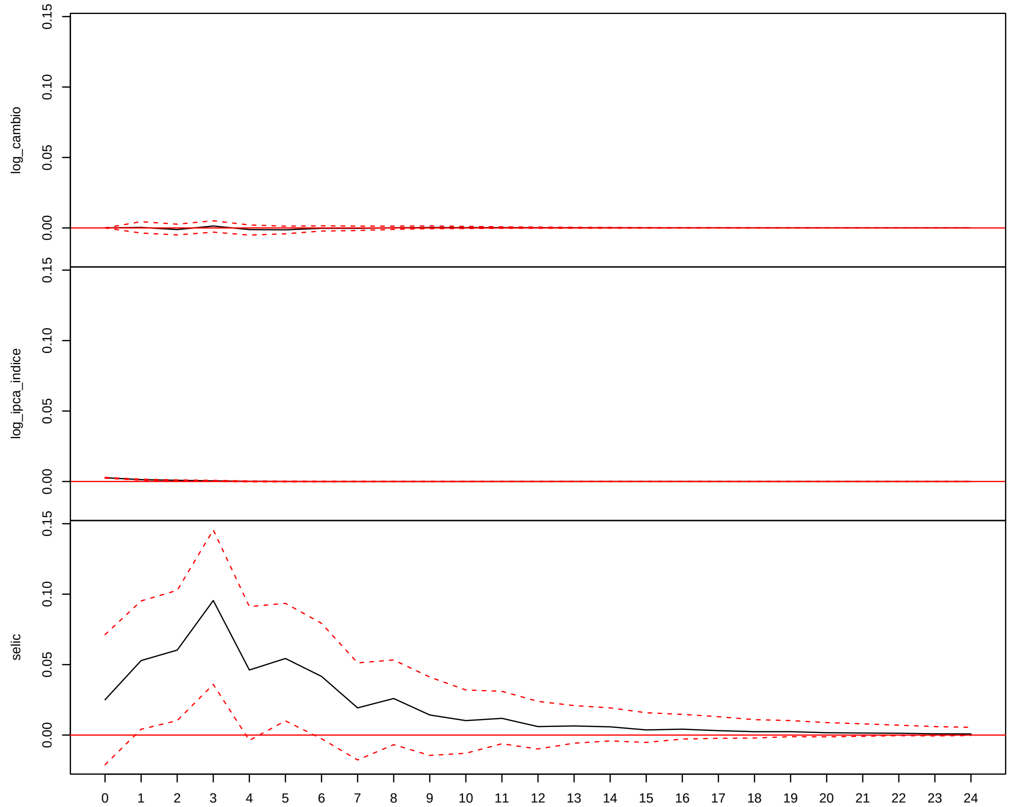
```
## ========================================
## selic = log_cambio.l1 + log_ipca_indice.l1 + selic.l1 + log_cambio.l2 + log_ipca_indice.l2 + selic.l2
##
##                    Estimate Std. Error t value    Pr(>|t|)
## log_cambio.l1      -1.12425    0.64377  -1.746     0.08194 .
## log_ipca_indice.l1  17.13762    7.95016   2.156     0.03204 *
## selic.l1            0.26925    0.06078   4.430 0.000013955 ***
## log_cambio.l2       1.83621    0.68366   2.686     0.00770 **
## log_ipca_indice.l2   5.79937    8.90484   0.651     0.51546
## selic.l2            0.32561    0.05959   5.464 0.000000109 ***
## log_cambio.l3      -0.88808    0.69077  -1.286     0.19973
## log_ipca_indice.l3  11.30763    8.93134   1.266     0.20663
## selic.l3            0.30656    0.05847   5.243 0.000000330 ***
## log_cambio.l4       0.12276    0.64267   0.191     0.84866
## log_ipca_indice.l4 -12.85473    7.95216  -1.617     0.10721
## selic.l4           -0.18488    0.05780  -3.199     0.00155 **
## const              -0.10735    0.05207  -2.062     0.04025 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.3486 on 258 degrees of freedom
## Multiple R-Squared: 0.5177,  Adjusted R-squared: 0.4953
## F-statistic: 23.08 on 12 and 258 DF,  p-value: < 0.00000000000000022
##
##
##
## Covariance matrix of residuals:
##                  log_cambio log_ipca_indice      selic
## log_cambio       0.00114238    -0.000011477 0.00001277
## log_ipca_indice -0.00001148     0.000007354 0.00006731
## selic            0.00001277     0.000067308 0.12153138
##
## Correlation matrix of residuals:
##                 log_cambio log_ipca_indice    selic
## log_cambio        1.000000         -0.1252 0.001083
## log_ipca_indice  -0.125223          1.0000 0.071198
## selic             0.001083          0.0712 1.000000
```

8. IRFs — duas ordenações

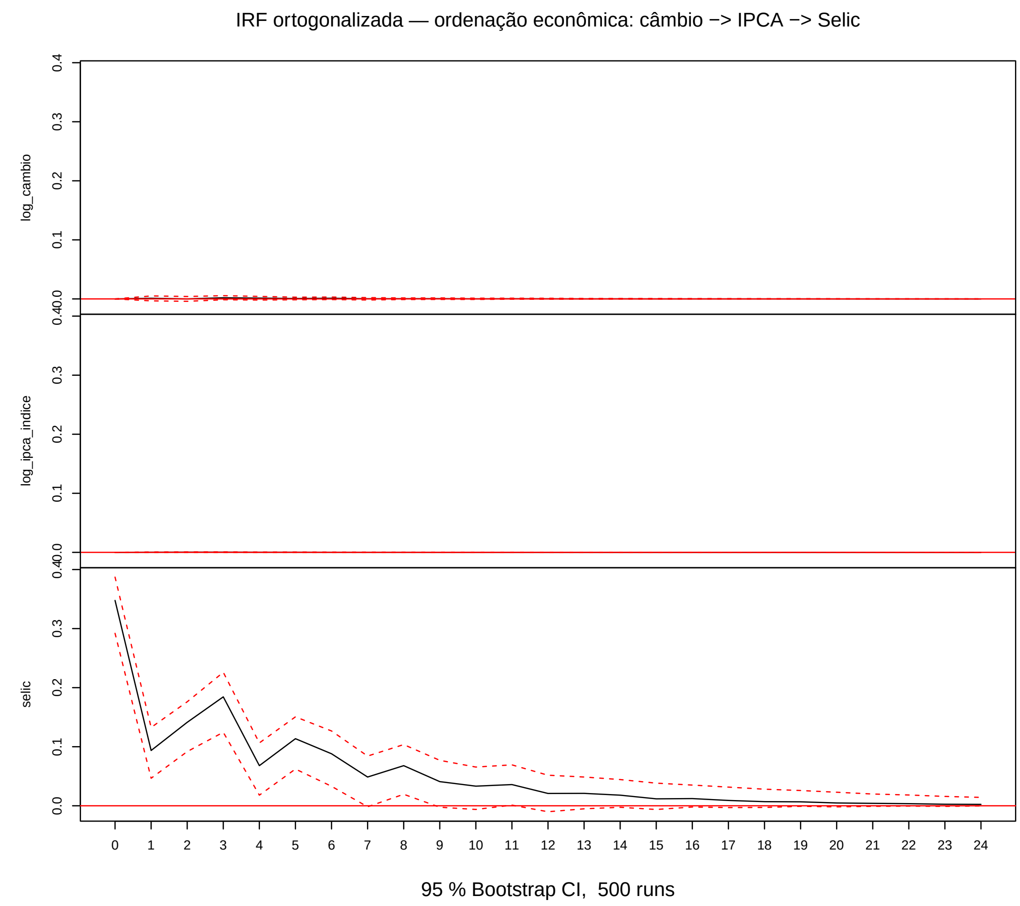
8.1 Ordenação econômica: câmbio → IPCA → Selic



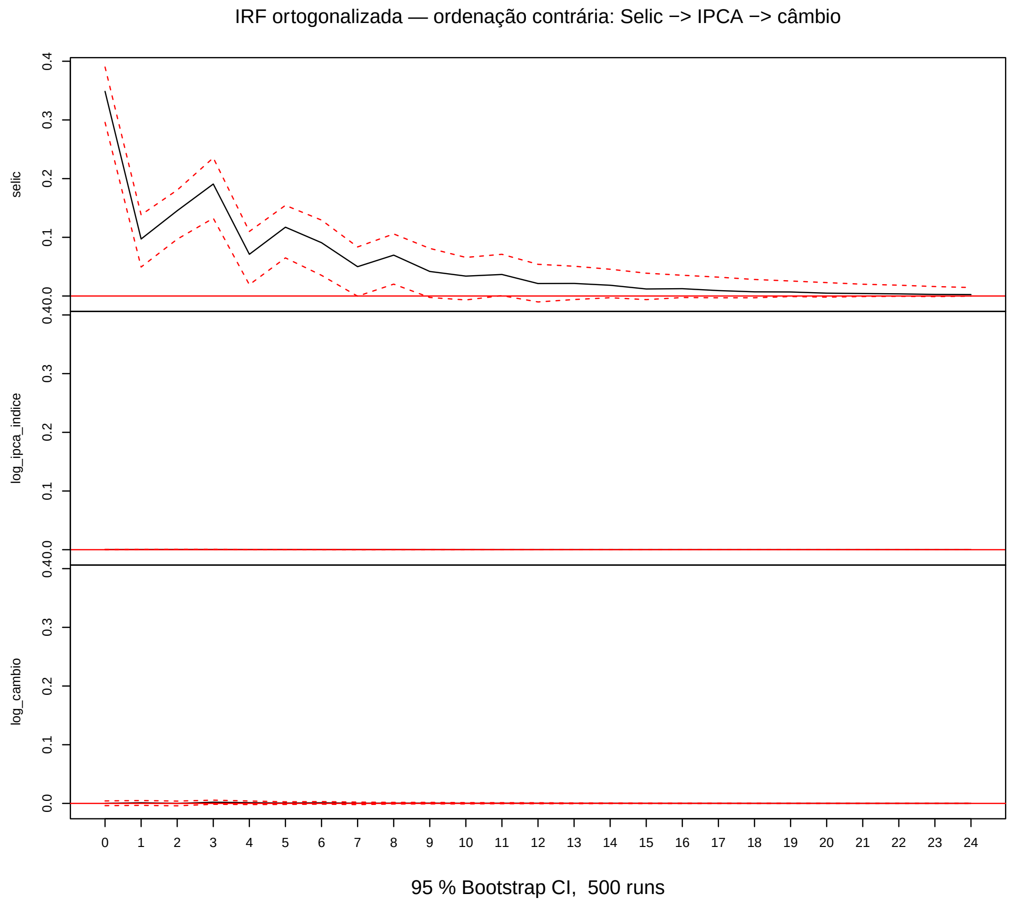
IRF ortogonalizada — ordenação econômica: câmbio −> IPCA −> Selic

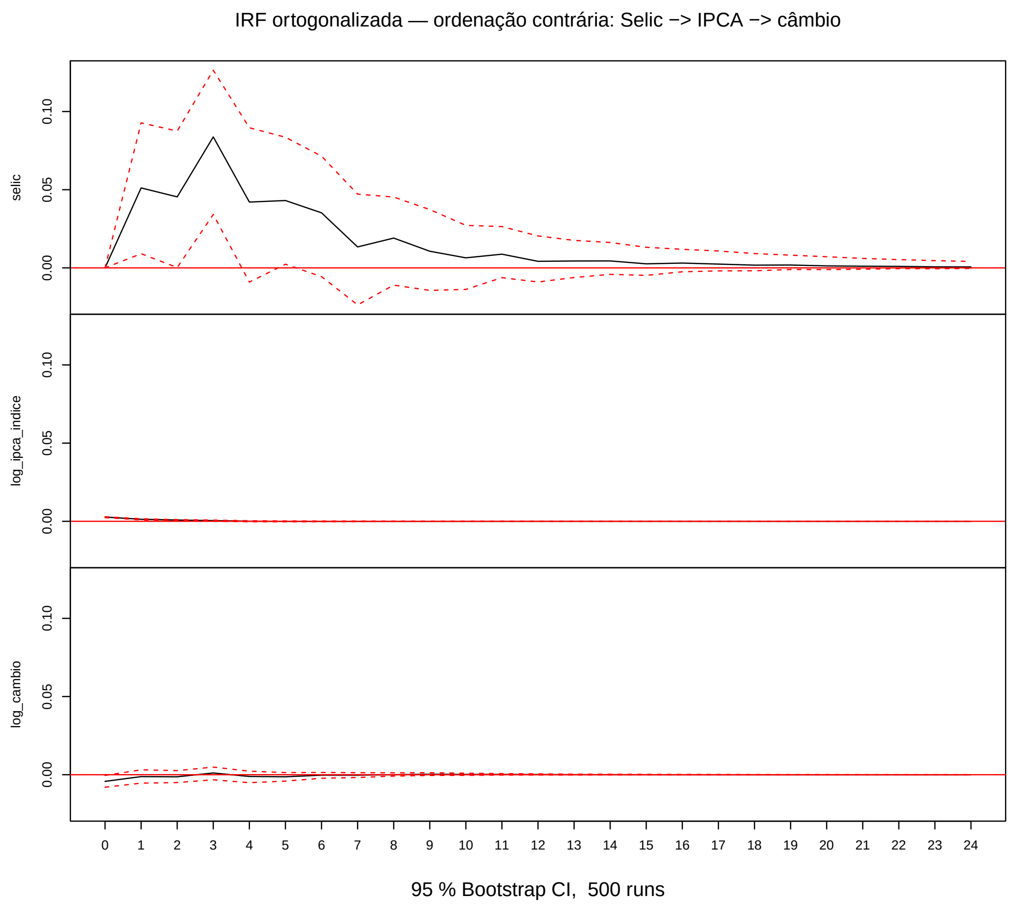


95 % Bootstrap CI, 500 runs

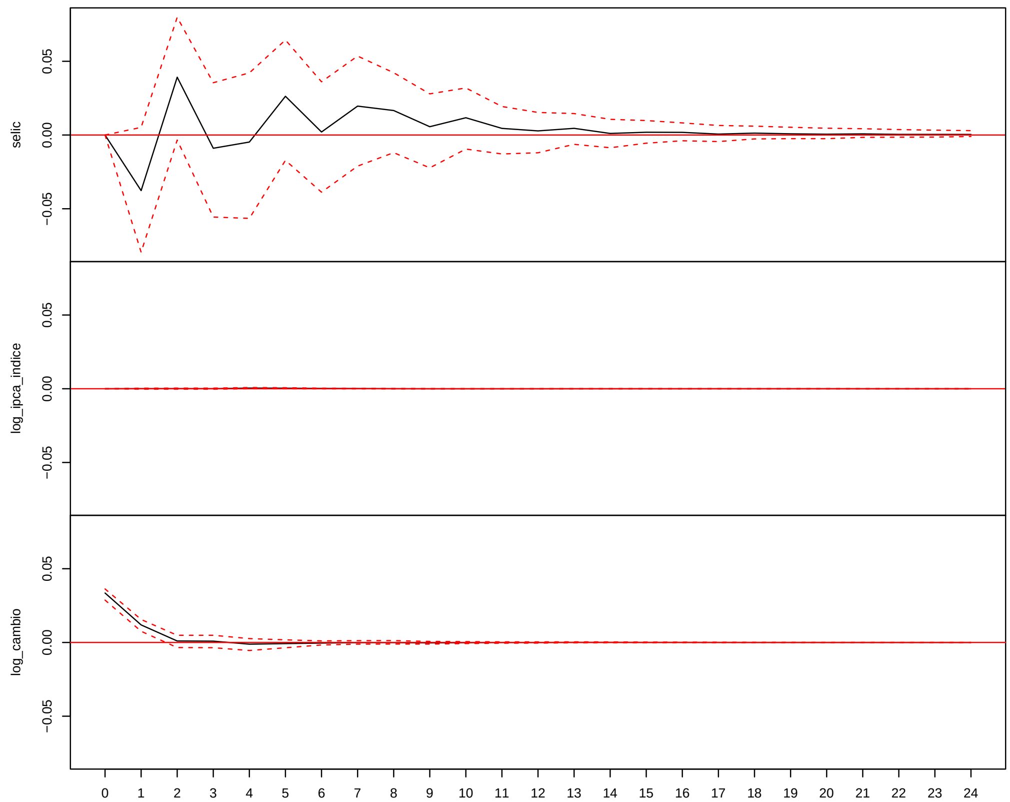

IRF ortogonalizada — ordenação econômica: câmbio −> IPCA −> Selic
log_cambio
log_ipca_indice
selic
0.0
0.1
0.2
0.3
0.4
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
95 % Bootstrap CI, 500 runs

8.2 Ordenação contrária: Selic → IPCA → câmbio




IRF ortogonalizada — ordenação contrária: Selic −> IPCA −> câmbio
selic
log_ipca_indice
log_cambio
0.00
0.05
0.10
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
95 % Bootstrap CI, 500 runs

IRF ortogonalizada — ordenação contrária: Selic -> IPCA -> câmbio



95 % Bootstrap CI, 500 runs

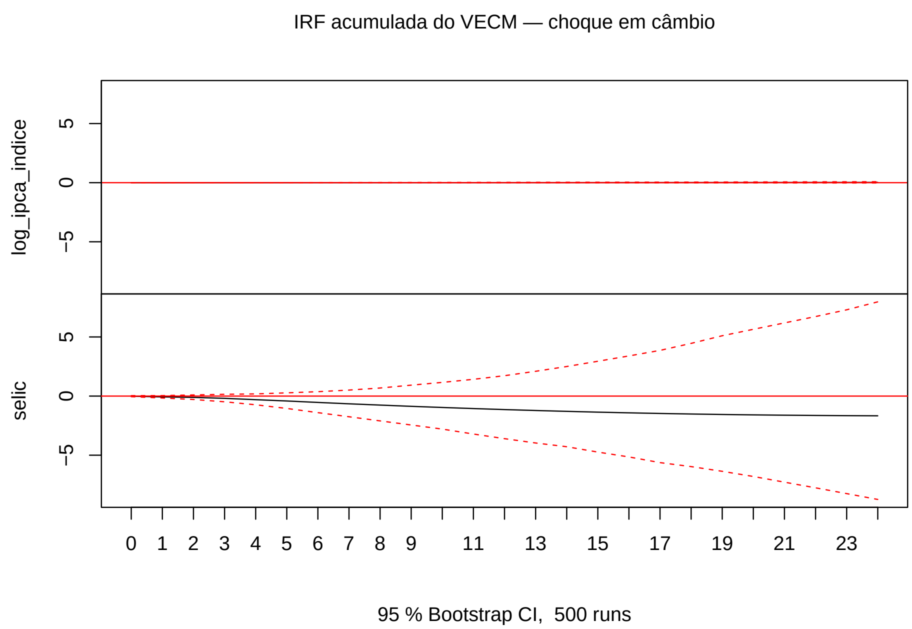
9. VECM

```
## Rank de cointegração adotado: 2

##
## --- Vetor(es) de cointegração - beta ---

##                          ect1     ect2
## log_ipca_indice.l5    1.0000   0.0000
## log_cambio.l5         0.0000   1.0000
## selic.l5              1.7769   3.5963
## constant            -24.5926 -29.9499

##
## --- Coeficientes de ajustamento - alpha ---

##      log_ipca_indice.d log_cambio.d selic.d
## ect1           -0.0014      -0.0026  0.0286
## ect2            0.0007       0.0011 -0.0200
```

```
## Response log_ipca_indice.d :
##
## Call:
## lm(formula = log_ipca_indice.d ~ ect1 + ect2 + log_ipca_indice.dl1 +
##     log_cambio.dl1 + selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 +
##     selic.dl2 + log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 +
##     log_ipca_indice.dl4 + log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0109938 -0.0013766 -0.0001643  0.0015508  0.0086590
##
## Coefficients:
##                       Estimate  Std. Error t value          Pr(>|t|)
## ect1                -0.00139193  0.00021547  -6.460 0.000000000521214 ***
## ect2                 0.00067998  0.00010297   6.604 0.000000000228526 ***
## log_ipca_indice.dl1  0.47270843  0.06188049   7.639 0.000000000000433 ***
## log_cambio.dl1       0.00271717  0.00499638   0.544            0.5870
## selic.dl1            0.00071122  0.00049187   1.446            0.1494
## log_ipca_indice.dl2  0.04405893  0.06894533   0.639            0.5234
## log_cambio.dl2       0.00211203  0.00531320   0.398            0.6913
## selic.dl2            0.00053739  0.00046887   1.146            0.2528
## log_ipca_indice.dl3 -0.03026812  0.06915062  -0.438            0.6620
## log_cambio.dl3      -0.00005437  0.00534824  -0.010            0.9919
## selic.dl3            0.00016597  0.00045411   0.365            0.7151
## log_ipca_indice.dl4 -0.09164536  0.06245038  -1.467            0.1435
## log_cambio.dl4       0.01326454  0.00500143   2.652            0.0085 **
## selic.dl4           -0.00036561  0.00045060  -0.811            0.4179
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002696 on 257 degrees of freedom
## Multiple R-squared:  0.7766, Adjusted R-squared:  0.7645
## F-statistic: 63.83 on 14 and 257 DF,  p-value: < 0.00000000000000022
##
##
## Response log_cambio.d :
##
## Call:
## lm(formula = log_cambio.d ~ ect1 + ect2 + log_ipca_indice.dl1 +
##     log_cambio.dl1 + selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 +
##     selic.dl2 + log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 +
##     log_ipca_indice.dl4 + log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.103236 -0.020492 -0.001125  0.018606  0.146363
##
## Coefficients:
##                       Estimate  Std. Error t value     Pr(>|t|)
## ect1                -0.00257704  0.00270040  -0.954        0.341
## ect2                 0.00105889  0.00129043   0.821        0.413
## log_ipca_indice.dl1  0.18794021  0.77552418   0.242        0.809
## log_cambio.dl1       0.34927894  0.06261765   5.578 0.0000000616 ***
```

```
## selic.dl1            0.00009176  0.00616445   0.015        0.988
## log_ipca_indice.dl2 -0.52599415  0.86406501  -0.609        0.543
## log_cambio.dl2      -0.10032473  0.06658829  -1.507        0.133
## selic.dl2           -0.00355544  0.00587620  -0.605        0.546
## log_ipca_indice.dl3  0.94479963  0.86663777   1.090        0.277
## log_cambio.dl3       0.04064460  0.06702746   0.606        0.545
## selic.dl3            0.00472859  0.00569118   0.831        0.407
## log_ipca_indice.dl4 -0.83391300  0.78266633  -1.065        0.288
## log_cambio.dl4      -0.05646625  0.06268093  -0.901        0.369
## selic.dl4            0.00034158  0.00564725   0.060        0.952
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03379 on 257 degrees of freedom
## Multiple R-squared:  0.1337, Adjusted R-squared:  0.08655
## F-statistic: 2.834 on 14 and 257 DF,  p-value: 0.0005689
##
##
## Response selic.d :
##
## Call:
## lm(formula = selic.d ~ ect1 + ect2 + log_ipca_indice.dl1 + log_cambio.dl1 +
##     selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 + selic.dl2 +
##     log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 + log_ipca_indice.dl4 +
##     log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.66054 -0.19978 -0.00839  0.18220  1.14967
##
## Coefficients:
##                     Estimate Std. Error t value   Pr(>|t|)
## ect1                 0.02861    0.02704   1.058    0.29108
## ect2                -0.02000    0.01292  -1.547    0.12301
## log_ipca_indice.dl1 16.45107    7.76611   2.118    0.03511 *
## log_cambio.dl1      -1.41805    0.62705  -2.261    0.02457 *
## selic.dl1            0.19450    0.06173   3.151    0.00182 **
## log_ipca_indice.dl2  6.12769    8.65276   0.708    0.47948
## log_cambio.dl2       1.52308    0.66682   2.284    0.02318 *
## selic.dl2            0.28156    0.05884   4.785 0.00000289 ***
## log_ipca_indice.dl3 12.44135    8.67853   1.434    0.15291
## log_cambio.dl3      -1.10362    0.67121  -1.644    0.10136
## selic.dl3            0.29013    0.05699   5.091 0.00000069 ***
## log_ipca_indice.dl4 -8.41328    7.83763  -1.073    0.28408
## log_cambio.dl4      -0.20335    0.62769  -0.324    0.74623
## selic.dl4           -0.15968    0.05655  -2.824    0.00512 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3384 on 257 degrees of freedom
## Multiple R-squared:  0.5508, Adjusted R-squared:  0.5263
## F-statistic: 22.51 on 14 and 257 DF,  p-value: < 0.0000000000000022
```

9.1 IRF acumulada do VECM via `vec2var`



10. Testes de especificação e diagnóstico

10.1 Diagnósticos do VAR

Table 8: Diagnósticos do VAR

Teste	Estatistica	p-valor	H0
Portmanteau ajustado — lags=12	88.4366	0.0914	Resíduos são ruído branco
Jarque-Bera multivariado	228.8536	0.0000	Resíduos normais
ARCH multivariado — lags=5	324.9408	0.0000	Ausência de efeitos ARCH
Breusch-Pagan — log_cambio	55.0185	0.0000	Homocedasticidade
Breusch-Pagan — log_ipca_indice	0.4253	0.8084	Homocedasticidade
Breusch-Pagan — selic	5.6938	0.0580	Homocedasticidade
Estabilidade — max raiz	0.8202	NA	Estável se max|raiz| <1

```
## --- Portmanteau VAR ---

##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 88.437, df = 72, p-value = 0.09141

##
## --- Jarque-Bera VAR ---
```

```
## $JB
##
##  JB-Test (multivariate)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 228.85, df = 6, p-value < 0.00000000000000022
##
##
## $Skewness
##
##  Skewness only (multivariate)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 30.866, df = 3, p-value = 0.000000907
##
##
## $Kurtosis
##
##  Kurtosis only (multivariate)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 197.99, df = 3, p-value < 0.00000000000000022

##
## --- ARCH VAR ---

##
##  ARCH (multivariate)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 324.94, df = 180, p-value = 0.0000000002085

##
## --- Breusch-Pagan VAR por equação ---

##
## log_cambio
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_var[, i]^2 ~ fitted_var[, i] + I(fitted_var[, i]^2))
## BP = 55.019, df = 2, p-value = 0.000000000001129
##
##
## log_ipca_indice
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_var[, i]^2 ~ fitted_var[, i] + I(fitted_var[, i]^2))
## BP = 0.42529, df = 2, p-value = 0.8084
##
##
## selic
```

```
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_var[, i]^2 ~ fitted_var[, i] + I(fitted_var[, i]^2))
## BP = 5.6938, df = 2, p-value = 0.05803
```

```
##
## --- Raízes VAR ---
```

```
##  [1] 0.8202 0.7444 0.7444 0.6874 0.6874 0.5953 0.5953 0.5540 0.5540 0.5126
## [11] 0.5126 0.4513
```

```
##
## Sistema VAR estável: TRUE
```

10.2 Diagnósticos do VECM

Table 9: Diagnósticos do VECM

Teste	Estatistica	p-valor	H0
Portmanteau ajustado — lags=12	90.1998	0.0256	Resíduos são ruído branco
Portmanteau robustez — K=6	89.7645	0.0037	Resíduos são ruído branco
Jarque-Bera multivariado	176.6925	0.0000	Resíduos normais
Breusch-Pagan — resids of log_ipca_indice	8.3064	0.0157	Homocedasticidade
Breusch-Pagan — resids of log_cambio	1.5087	0.4703	Homocedasticidade
Breusch-Pagan — resids of selic	30.4995	0.0000	Homocedasticidade
Estabilidade — max raiz	1.0000	NA	Estável se max|raiz| $<$ 1

```
## --- Portmanteau VECM ---
```

```
##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object vecm_as_var
## Chi-squared = 90.2, df = 66, p-value = 0.02562
```

```
##
## --- Portmanteau VECM robustez K=6 ---
```

```
##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object vecm6_as_var
## Chi-squared = 89.764, df = 57, p-value = 0.003652
```

```
##
## --- Jarque-Bera VECM ---
```

```
## $JB
##
##  JB-Test (multivariate)
##
## data:  Residuals of VAR object vecm_as_var
## Chi-squared = 176.69, df = 6, p-value < 0.00000000000000022
##
##
## $Skewness
##
##  Skewness only (multivariate)
##
## data:  Residuals of VAR object vecm_as_var
## Chi-squared = 22.359, df = 3, p-value = 0.00005492
##
##
## $Kurtosis
##
##  Kurtosis only (multivariate)
##
## data:  Residuals of VAR object vecm_as_var
## Chi-squared = 154.33, df = 3, p-value < 0.00000000000000022
```

```
##
## --- Breusch-Pagan VECM por equação ---
```

```
##
## resids of log_ipca_indice
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_vecm[, i]^2 ~ fitted_vecm[, i] + I(fitted_vecm[, i]^2))
## BP = 8.3064, df = 2, p-value = 0.01571
##
##
## resids of log_cambio
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_vecm[, i]^2 ~ fitted_vecm[, i] + I(fitted_vecm[, i]^2))
## BP = 1.5087, df = 2, p-value = 0.4703
##
##
## resids of selic
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_vecm[, i]^2 ~ fitted_vecm[, i] + I(fitted_vecm[, i]^2))
## BP = 30.5, df = 2, p-value = 0.0000002383
```

```
##
## --- Raízes VECM ---
```

```
##  [1] 1.0000 0.9983 0.9171 0.9171 0.7307 0.7307 0.6611 0.6611 0.6034 0.6034
```

```
## [11] 0.5421 0.5421 0.5221 0.5221 0.3841
```

```
##
## Sistema VECM com todas as raízes < 1: FALSE
```